PRACTICAL – 12

1. Aim

Simulate door lock by interfacing fingerprint sensor and relay module with raspberry Pi. (This practical instead of door, bulb is used to show the output).

2. Prior Concepts

i) Basic Raspberry Pi architecture and GPIO programming.

ii) Knowledge of UART serial communication.

iii) Basics of biometric authentication.

iv) Relay module operation and switching principles.

3. New Concepts

i) Interfacing a biometric fingerprint sensor with Raspberry Pi.

ii) Controlling a relay based on authentication results.

iii) Integration of sensing, processing, and actuation in an embedded system.

4. Hardware Required

i) Fingerprint Sensor (R307, AS608, etc.)

ii) Relay Module (5V)

iii) Bulb

iv) Jumper Wires

5. Procedure

Pin Connection

i) Fingerprint Sensor → Raspberry Pi

Fingerprint Sensor	Raspberry Pi
VCC	5V
GND	GND
TX	GPIO15(RXD)
RX	GPIO14(TXD)

ii) Relay Module to Raspberry Pi

Relay Module	Raspberry Pi
VCC	5V
GND	GND
IN	GPIO18

iii) Bulb Connection

Bulb	Relay Module
Positive Terminal	No (Normally Open)
Negative Terminal	GND/Power Return

Input IMAGE

i) Install Python3 on Raspi.

Write the below code in command prompt:

sudo apt install idle3

Enable UART on Raspberry Pi:

sudo raspi-config

Navigate to:

Interface Options → Serial Port

Login Shell? → No

Serial Hardware? → Yes

Reboot:

sudo reboot

Install Libraries

sudo apt update

sudo apt install python3-pip -y

pip3 install pyfingerprint

6. Implementation with Python Code

Code:

```
from pyfingerprint.pyfingerprint import PyFingerprint

import RPi.GPIO as GPIO

from time import sleep

# Relay connected to GPIO18
RELAY_PIN = 18

GPIO.setwarnings(False)
GPIO.setmode(GPIO.BCM)

GPIO.setup(RELAY_PIN, GPIO.OUT)
```

```
# Initially bulb OFF
GPIO.output(RELAY_PIN, GPIO.LOW)

# Initialize Fingerprint Sensor
try:

    fp = PyFingerprint(
        '/dev/serial0',
        57600,
        0xFFFFFFFF,
        0x00000000
    )

    if not fp.verifyPassword():
        raise ValueError('Fingerprint sensor password incorrect')

except Exception as e:

    print("Fingerprint Sensor Initialization Failed")

    print(e)

    exit(1)

print("= = = =
```

```
    print("\nPlace Finger on Sensor...")

    # Wait until finger is placed
    while not fp.readImage():
        pass

    # Convert image to characteristics
    fp.convertImage(0x01)

    # Search fingerprint template
    result = fp.searchTemplate()

positionNumber = result[0]

if positionNumber >= 0:

    print("Fingerprint Matched")
    print("Bulb ON")

    # Turn ON Bulb through Relay
    GPIO.output(RELAY_PIN, GPIO.HIGH)

    # Keep bulb ON for 5 seconds
    sleep(5)

    # Turn OFF Bulb
    GPIO.output(RELAY_PIN, GPIO.LOW)

    print("Bulb OFF")

else:

    print("Unauthorized User")
    print("Access Denied")
```

```
    print("Bulb Remains OFF")

    GPIO.output(RELAY_PIN, GPIO.LOW)

    sleep(2)
```

7. Result:

PRACTICAL – 13

1. Aim

Implement basic Publisher – Subscriber using MQTT. Publish messages using mqtt. Subscribe and control LED.

2. Prior Concepts

i) Basic Raspberry Pi architecture and GPIO programming.

ii) Client–Server communication.

iii) Command-line operations in Linux.

3. New Concepts

i) MQTT publish-subscribe communication model.

ii) Working of MQTT broker, publisher and subscriber.

iii) Using Paho MQTT library in Python.

iv) Event-driven programming using MQTT callbacks.

v) Building cloud-ready IoT applications.

4. Hardware Required